

LEVEL 4 · WRITE-UP

Planet Hunt

Every clue on the slip is a filterable fact. The fastest route is the NASA Exoplanet Archive, but a plain search for the nickname gets there too.

Identifying the planet

The combination is unique: a super-Earth (~ 2 Earth radii, ~ 8 Earth masses), an ~ 18 -hour orbital period, carbon-rich, around a Sun-like star ~ 40 ly away in Cancer, discovered 2004. In the Exoplanet Archive, filtering on orbital period < 1 day and distance ≈ 40 ly in Cancer narrows it to one well-known world:

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Designation : 55 Cancr i e    (host star 55 Cancr i A / Rho^1 Cancr i)
Type        : super-Earth, ~2 R_Earth, ~8 M_Earth
Orbit       : ~0.74 days    (about 18 hours)
Distance    : ~40 light-years, constellation Cancer
Discovered  : 2004 (innermost of the 55 Cancr i planets)
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The carbon-rich clue is the giveaway for the nickname: the press widely dubbed 55 Cancr i e “the diamond planet.” Searching that nickname lands on the same planet.

Recovering the proper name

The challenge asks for the IAU proper name, not the designation or nickname. In the IAU's NameExoWorlds naming, 55 Cancr i e was officially named Janssen (after the spectacle-maker associated with the invention of the telescope; its host star was named Copernicus). This is listed on the planet's NASA Exoplanet Archive / Wikipedia entry under “proper name.”

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CTC{janssen}
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Builder notes

- Make the flag case-insensitive in CTFd so Janssen is accepted.
- Add a short hint that costs points: “The press called it the diamond planet. The IAU gave it a human name.”
- Optionally reject the obvious near-misses in the description itself (55cancr i e / diamond) so solvers know those are not the flag.
- This level is self-contained — it does not depend on Level 3's answer. Its only job in the chain is to unlock Level 5.